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FACULTY OF INFORMATION TECHNOLOGY 1



SUBJECT : BI

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Chapter 1: Problem Introduction

I. Project Objectives

This project aims to achieve the following core objectives:

- **Comprehensive Analysis of Transactional Data:** The project analyzes transaction data from an e-commerce system to identify best-selling products, understand customer purchasing behavior, and evaluate revenue performance across different time periods.
- **Support for Strategic Decision-Making:** The project provides the Sales and Marketing departments with accurate, data-driven insights presented through interactive dashboards and business intelligence reports. This approach minimizes decision-making based on intuition and improves strategic planning.
- **Optimization of Business Operations and Supply Chain Management:** By identifying recurring purchasing patterns and seasonal sales trends, the project supports inventory optimization, targeted promotional campaigns, and enhanced customer relationship management. These insights help improve operational efficiency and overall business performance.

II. Motivation and Significance

In the digital era, e-commerce platforms generate massive volumes of transactional data every day. When data remains fragmented and underutilized, organizations lose valuable opportunities to gain competitive advantages. Transforming raw data into actionable business intelligence has become a critical business requirement.

2.1 Enterprise-Wide Data Transparency

The project establishes a centralized analytical platform that eliminates information silos between departments such as Logistics, Marketing, Finance, and Sales. All stakeholders access a single source of truth, ensuring consistency and transparency in performance evaluation and financial reporting.

2.2 Inventory Cost Optimization

The analysis enables the identification of fast-moving and slow-moving products. This information supports more accurate inventory planning, reduces storage costs, minimizes overstock situations, and improves supply chain efficiency.

2.3 Improved Customer Retention

Understanding customer purchasing behavior allows businesses to develop more effective retention strategies. Previous studies indicate that increasing customer retention rates by a small percentage can significantly improve profitability while reducing customer acquisition costs.

2.4 Automated and Standardized Reporting

The project replaces manual spreadsheet-based reporting processes with an automated Business Intelligence solution. This ensures data consistency, reduces reporting time, and enables real-time monitoring of key business metrics through interactive dashboards.

Chapter 2: Dataset Description

I. Data Source

The primary dataset used in this project is the Online Retail Dataset stored in CSV format: Online Retail.csv. This dataset contains real-world transactional records collected from a UK-based online retail company.

1.1 Data Collection Period

The dataset records transactions that occurred between **2010 and 2011**, providing historical sales information over a two-year period.

1.2. Original Dataset Attributes

The most important fields contained in the raw dataset include:

Attribute	Description
InvoiceNo	Unique invoice number representing a customer order
StockCode	Product identifier
Description	Product description
Quantity	Number of units purchased
InvoiceDate	Date and time of the transaction
UnitPrice	Price per product unit
CustomerID	Unique customer identifier
Country	Customer's country of origin

These attributes provide the foundation for building the data warehouse, generating dimension tables, constructing the fact table, and performing multidimensional business analysis.

1.3 Dataset Characteristics

The Online Retail dataset represents a typical transactional sales system and contains information related to:

- Customer purchasing activities
- Product sales performance
- Revenue generation
- Geographic distribution of customers
- Temporal sales trends

These characteristics make the dataset highly suitable for implementing a Business Intelligence solution based on a Data Warehouse architecture and Star Schema modeling.

II. Data Schema

To support multidimensional storage and analysis, the raw e-commerce data is restructured into a data warehouse consisting of one central fact table and four dimension tables.

2.1 fact_orders — Central Fact Table

The fact_orders table stores detailed transaction records and numerical measures used for analysis.

Columns:

- fact_id: Primary key
- invoice_no: Invoice number
- customer_key: Foreign key referencing dim_customer
- product_key: Foreign key referencing dim_product
- time_key: Foreign key referencing dim_time
- region_key: Foreign key referencing dim_region
- quantity: Quantity of products purchased
- unit_price: Unit price of each product
- revenue: Total revenue of the transaction

Calculation formula: $\text{revenue} = \text{quantity} \times \text{unit_price}$

2.2 dim_customer — Customer Dimension

The dim_customer table stores cleaned and standardized customer information.

Columns:

- customer_key: Primary key
- customer_id: customer_id: Original customer identifier or generated guest customer identifier

- country: Customer's country

2.3 dim_product — Product Dimension

The dim_product table stores product information for product-level analysis.

Columns:

- product_key: Primary key
- stock_code: Product code
- description: Product description

2.4 dim_time — Time Dimension

The dim_time table is used to support time-based analysis such as revenue by day, month, quarter, and year.

Columns:

- time_key: Primary key
- invoice_date: Transaction date
- day: Day
- month: Month
- quarter: Quarter
- year: Year

2.5 dim_region — Region Dimension

The dim_region table stores geographical information and groups countries into broader regions.

Columns:

- region_key: Primary key
- country: Country name
- region_group: Region group, such as UK, Europe, North America, Oceania, or Other Internat

III. Preliminary Statistics and Data Profiling

Before executing the ETL process, an initial data profiling stage was performed to evaluate data quality, identify inconsistencies, and understand the overall characteristics of the Online Retail dataset.

3.1 Dataset Overview

The original dataset contains:

- Total Transaction Records: 541,909 rows
- Business Domain: Online Retail / E-commerce
- Transaction Period: December 2010 to December 2011
- Data Source: UK-based online retail company

The dataset includes detailed information about customer purchases, product identifiers, transaction dates, quantities, prices, and customer locations.

3.2 Customer and Product Characteristics

The dataset contains thousands of unique customers and products, making it suitable for multidimensional business analysis.

After the ETL process and dimensional modeling, the warehouse supports analytical reporting across multiple business perspectives, including:

- Customer behavior analysis
- Product performance analysis
- Geographic analysis
- Time-series revenue analysis

3.3 Geographic Distribution

The United Kingdom represents the dominant market within the dataset and contributes the majority of transaction records.

Additional transactions originate from multiple international markets, including:

- Germany
- France
- EIRE (Ireland)
- Netherlands
- Belgium
- Australia
- United States
- Canada
- Other international countries

To simplify analytical reporting, countries are grouped into broader business regions such as:

- UK
- Europe
- North America
- Oceania
- Other International

This regional classification is later used in the Region Dimension and Power BI dashboards.

3.4 Data Quality Issues Identified

Several data quality issues were identified during the profiling phase and subsequently addressed during the transformation stage.

3.4.1 Negative Quantity Values

A significant number of records contain negative Quantity values.

These records generally represent:

- Order cancellations
- Product returns
- Refund transactions

Since the objective of this project is sales and revenue analysis, these records are excluded from the analytical warehouse.

3.4.2 Zero or Negative Unit Prices

Some transactions contain UnitPrice values equal to or below zero.

These records may represent:

- Promotional gifts
- Free products
- Data entry errors
- System adjustment records

Because such transactions do not generate actual sales revenue, they are removed during data cleansing.

3.4.3 Missing Customer Identifiers

Some transactions contain missing CustomerID values. Instead of discarding these records and losing valuable revenue information, the ETL process generates anonymous customer identifiers based on the customer's country.

Examples include:

- United Kingdom_Guest
- France_Guest
- Germany_Guest

This approach preserves transactional revenue while maintaining customer-level analytical capabilities.

3.4.4 Missing Product Descriptions

Missing product descriptions, if any, are replaced with the value: No Description. This ensures completeness and consistency within the Product Dimension table.

3.4.5 Data Profiling Outcome

The profiling process confirmed that the dataset is suitable for Business Intelligence and Data Warehouse implementation after cleansing and standardization.

All identified quality issues were addressed through the ETL process, resulting in a clean, consistent, and analysis-ready dataset.

The transformed data is subsequently organized into a Star Schema consisting of:

- Fact Orders
- Customer Dimension
- Product Dimension
- Time Dimension
- Region Dimension

This structure provides the foundation for multidimensional analysis and interactive dashboard development in Power BI.

Chapter 3: Data Warehouse Design

I. Data Model

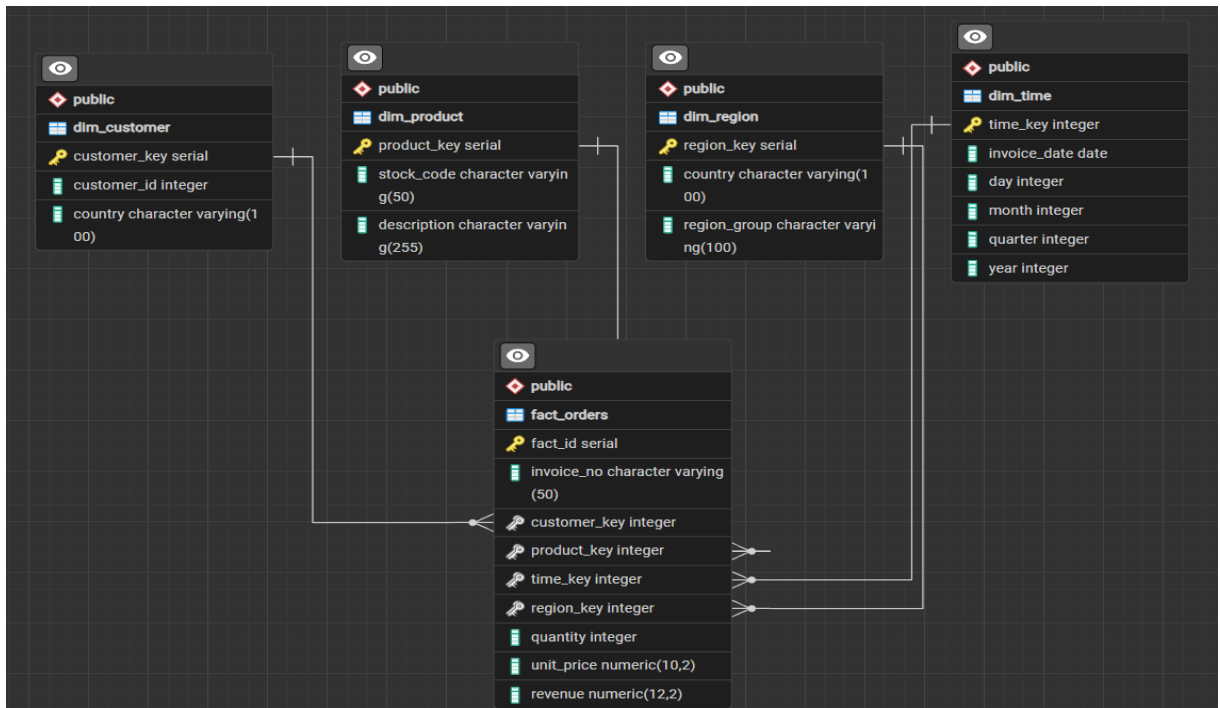
The system adopts a Star Schema architecture, which is one of the most widely used data warehouse models for analytical processing (OLAP). In this design, the fact_orders table serves as the central fact table and is directly connected to four dimension tables: dim_customer, dim_product, dim_time, and dim_region.

This structure simplifies analytical queries, improves reporting performance, and provides a clear business-oriented view of the data.

II. Relationship Structure (Many-to-One Relationships)

The relationships between the central fact table and the dimension tables follow a Many-to-One (N:1) pattern. The fact_orders table stores foreign keys, while each dimension table contains unique primary keys.

- fact_orders (Many) → dim_customer (One) through the customer_key foreign key. A single customer can generate multiple transactions over time.
- fact_orders (Many) → dim_product (One) through the product_key foreign key. A product may appear in many different transactions.
- fact_orders (Many) → dim_time (One) through the time_key foreign key. Multiple transactions can occur on the same date and be aggregated by day, month, quarter, or year.
- fact_orders (Many) → dim_region (One) through the region_key foreign key. Multiple transactions may belong to the same country or regional group.



III. Reasons for Choosing the Star Schema

3.1 High Query Performance

The Star Schema minimizes the number of joins required during analytical queries compared to more normalized models such as the Snowflake Schema. This allows Power BI dashboards to return results quickly, even when processing large datasets.

3.2 Simplicity and Readability

The structure is intuitive and closely aligned with business analysis requirements. Users can easily analyze revenue and sales performance from different perspectives, such as customer, product, time, and region.

3.3 Scalability

The model can be extended easily when new analytical requirements arise. Additional dimensions, such as Marketing Channel, Sales Representative, or Payment Method, can be added without significantly changing the existing warehouse structure.

Chapter 4: ETL Process (Extract – Transform – Load)

I. Extract

The extraction phase is responsible for collecting raw transactional data from the Online Retail dataset and preparing it for subsequent processing stages.

The system reads the source dataset directly from the CSV file Online Retail.csv using the Pandas library in Python. Before loading the file, the ETL process verifies whether the source file exists in the specified directory. If the file cannot be found, the system immediately stops execution and raises an exception to prevent incomplete or incorrect data loading.

The extracted dataset contains detailed e-commerce transaction records, including invoice numbers, product identifiers, customer identifiers, purchase quantities, unit prices, transaction dates, and customer countries.

After extraction, the dataset is loaded into memory as a Pandas DataFrame, which serves as the staging area for all transformation operations.

II. Transform

The transformation phase performs data cleansing, standardization, enrichment, and dimensional modeling to ensure data quality and consistency before loading into the data warehouse.

2.1. Data Quality Filtering

To maintain analytical accuracy, invalid transactional records are removed.

The system automatically excludes:

- Transactions with $\text{Quantity} \leq 0$
- Transactions with $\text{UnitPrice} \leq 0$

These records typically represent canceled orders, returned products, system errors, or transactions that do not contribute to actual sales revenue.

This filtering process ensures that all remaining records represent valid revenue-generating sales activities.

2.2 Missing Data Handling

Missing values are handled according to business requirements.

Records with missing:

- InvoiceDate
- Country

are removed because these fields are mandatory for time-based and region-based analysis.

For missing CustomerID values, transactions are not discarded. Instead, anonymous customer identifiers are generated dynamically using the customer's country information. Examples:

- United Kingdom_Guest
- France_Guest
- Germany_Guest

This approach preserves sales revenue while allowing guest customers to remain traceable during analysis.

2.3 Data Standardization

Several fields are standardized before dimensional modeling:

- CustomerID is converted to string format.
- StockCode is converted to string format.
- InvoiceDate is converted to a DateTime data type.
- Revenue is calculated using: $\text{Revenue} = \text{Quantity} \times \text{UnitPrice}$

The Revenue column becomes the primary financial measure used throughout the analytical dashboards.

2.4. Customer Dimension Construction

- The customer dimension is created using unique customer identifiers. To prevent duplicate customer records, only one record is retained for each CustomerID.
- The transformation process applies: `drop_duplicates(subset = ['CustomerID'], keep='last')`. This ensures that every customer appears exactly once within the dimension table
- A surrogate key named `customer_key` is generated and assigned sequentially.

2.5 Product Dimension Construction

- The product dimension is created using unique StockCode values.

- In real-world retail data, the same product code may appear with multiple product descriptions due to spelling corrections, formatting differences, or manual updates.
- To prevent duplicate dimension members and fact table duplication during joins, the ETL process keeps only one record per StockCode.
- The transformation process applies:
`drop_duplicates(subset=['StockCode'], keep='last')`
- Missing product descriptions are replaced with: No Description
- A surrogate key named `product_key` is generated for each product.

2.6 Time Dimension Construction

The time dimension is generated automatically from transaction dates. For every unique transaction date, the following analytical attributes are created:

- TimeKey
- Day
- Month
- Quarter
- Year

The TimeKey follows the format YYYYMMDD. This structure supports temporal analysis at multiple aggregation levels.

2.7 Region Dimension Construction

The region dimension is derived from unique country values. Countries are grouped into broader geographical categories through a predefined mapping table.

Examples:

- United Kingdom → UK
- Germany → Europe
- France → Europe
- Australia → Oceania
- USA → North America
- Canada → North America

Countries not explicitly mapped are assigned to: Other International

A surrogate key named `region_key` is generated for each country.

2.8 Foreign Key Mapping

After all dimension tables have been created, the ETL process retrieves surrogate keys from PostgreSQL and maps them back into the transactional dataset.

The following foreign keys are assigned:

- customer_key
- product_key
- region_key
- time_key

This process establishes the relationships required by the Star Schema.

2.9 Fact Table Construction

The fact table stores quantitative business measures and references to all dimension tables. The final fact table contains:

- invoice_no
- customer_key
- product_key
- time_key
- region_key
- quantity
- unit_price
- revenue

This table becomes the central component of the data warehouse.

III. Load

The load phase transfers the transformed data into PostgreSQL. Before loading new data, the ETL process removes existing warehouse tables. This ensures that every ETL execution generates a clean and consistent warehouse environment without duplicate records.

The dimension tables are loaded first:

- dim_customer
- dim_product
- dim_time
- dim_region

After all dimensions have been successfully loaded and surrogate keys have been generated, the fact table is populated. The loading sequence follows standard data warehouse practices:

- Dimensions First
- Foreign Key Mapping
- Fact Table

The final PostgreSQL database serves as the centralized analytical repository that is subsequently connected to Power BI for dashboard development, reporting, and business intelligence analysis.

IV. ETL Scripts

The ETL implementation is developed in Python using Pandas and SQLAlchemy. The script performs the complete ETL workflow, including:

- Data extraction from CSV files
- Data cleansing and validation
- Missing value handling
- Guest customer generation
- Revenue calculation
- Dimension table construction
- Surrogate key generation
- Foreign key mapping
- Fact table creation
- PostgreSQL loading

```

from pathlib import Path

import pandas as pd
from sqlalchemy import create_engine, text

USER = 'postgres'
PASSWORD = '123456'
HOST = 'localhost'
PORT = '5432'
DATABASE = 'ecommerce_dw_ver1'

BASE_DIR = Path(__file__).resolve().parent
CSV_PATH = BASE_DIR / 'data' / 'Online Retail.csv'

engine =
create_engine(f"postgresql+psycopg2://{USER}:{PASSWORD}@{HOST}:{PORT}/{DATABASE}")

print("--- Dang doc du lieu ---")
if not CSV_PATH.exists():
    raise FileNotFoundError(f"Khong tim thay file du lieu: {CSV_PATH}")

df = pd.read_csv(CSV_PATH, encoding='ISO-8859-1')

print("--- Dang lam sach va bien doi du lieu ---")

df = df[(df['Quantity'] > 0) & (df['UnitPrice'] > 0)].copy()
df = df.dropna(subset=['InvoiceDate', 'Country'])

df['CustomerID'] = df['CustomerID'].fillna(df['Country'] + '_Guest')
df['CustomerID'] = df['CustomerID'].astype(str)

df['StockCode'] = df['StockCode'].astype(str)
df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
df['Revenue'] = df['Quantity'] * df['UnitPrice']

dim_customer_df = (
    df[['CustomerID', 'Country']]
    .drop_duplicates(subset=['CustomerID'], keep='last')
    .reset_index(drop=True)
)
dim_customer_df.insert(0, 'customer_key', range(1, len(dim_customer_df) + 1))

dim_product_df = (
    df[['StockCode', 'Description']]

```

```

        .drop_duplicates(subset=['StockCode'], keep='last')
        .reset_index(drop=True)
    )
    dim_product_df['Description'] = dim_product_df['Description'].fillna('No
Description')
    dim_product_df.insert(0, 'product_key', range(1, len(dim_product_df) + 1))

    dim_time_df = pd.DataFrame({'InvoiceDate':
df['InvoiceDate'].dt.normalize().unique()})
    dim_time_df['TimeKey'] =
dim_time_df['InvoiceDate'].dt.strftime('%Y%m%d').astype(int)
    dim_time_df['Day'] = dim_time_df['InvoiceDate'].dt.day
    dim_time_df['Month'] = dim_time_df['InvoiceDate'].dt.month
    dim_time_df['Quarter'] = dim_time_df['InvoiceDate'].dt.quarter
    dim_time_df['Year'] = dim_time_df['InvoiceDate'].dt.year

    dim_region_df = df[['Country']].drop_duplicates().reset_index(drop=True)
    region_map = {
        'United Kingdom': 'UK',
        'Germany': 'Europe',
        'France': 'Europe',
        'EIRE': 'Europe',
        'Norway': 'Europe',
        'Australia': 'Oceania',
        'USA': 'North America',
        'Canada': 'North America',
    }
    dim_region_df['region_group'] =
dim_region_df['Country'].map(region_map).fillna('Other International')
    dim_region_df.insert(0, 'region_key', range(1, len(dim_region_df) + 1))

    print("--- Dang nap du lieu vao Database ---")

    with engine.begin() as conn:
        for table_name in ['fact_orders', 'dim_region', 'dim_time',
'dim_product', 'dim_customer']:
            conn.execute(text(f'DROP TABLE IF EXISTS {table_name} CASCADE'))

    dim_customer_df.columns = ['customer_key', 'customer_id', 'country']
    dim_product_df.columns = ['product_key', 'stock_code', 'description']
    dim_time_df.columns = ['invoice_date', 'time_key', 'day', 'month',
'quarter', 'year']
    dim_region_df.columns = ['region_key', 'country', 'region_group']

    dim_customer_df.to_sql('dim_customer', engine, if_exists='replace',
index=False)
    dim_product_df.to_sql('dim_product', engine, if_exists='replace',

```

```

index=False)
dim_time_df.to_sql('dim_time', engine, if_exists='replace', index=False)
dim_region_df.to_sql('dim_region', engine, if_exists='replace',
index=False)

print("--- Dang anh xa khoa ngoai cho bang Fact ---")

db_cust = pd.read_sql('SELECT customer_key, customer_id FROM dim_customer',
engine)
db_prod = pd.read_sql('SELECT product_key, stock_code FROM dim_product',
engine)
db_reg = pd.read_sql('SELECT region_key, country FROM dim_region', engine)

db_cust['customer_id'] = db_cust['customer_id'].astype(str)
db_prod['stock_code'] = db_prod['stock_code'].astype(str)

df = df.merge(db_cust, left_on='CustomerID', right_on='customer_id',
how='left')
df = df.merge(db_prod, left_on='StockCode', right_on='stock_code',
how='left')
df = df.merge(db_reg, left_on='Country', right_on='country', how='left')

df['time_key'] = df['InvoiceDate'].dt.strftime('%Y%m%d').astype(int)

fact_orders_df = df[
    [
        'InvoiceNo',
        'customer_key',
        'product_key',
        'time_key',
        'region_key',
        'Quantity',
        'UnitPrice',
        'Revenue',
    ]
].copy()

fact_orders_df.columns = [
    'invoice_no',
    'customer_key',
    'product_key',
    'time_key',
    'region_key',
    'quantity',
    'unit_price',
    'revenue',
]

```

```
print("--- Dang nap bang fact_orders vao PostgreSQL ---")
fact_orders_df.to_sql('fact_orders', engine, if_exists='replace',
index=False)

print("ETL hoan tat thanh cong!")
```

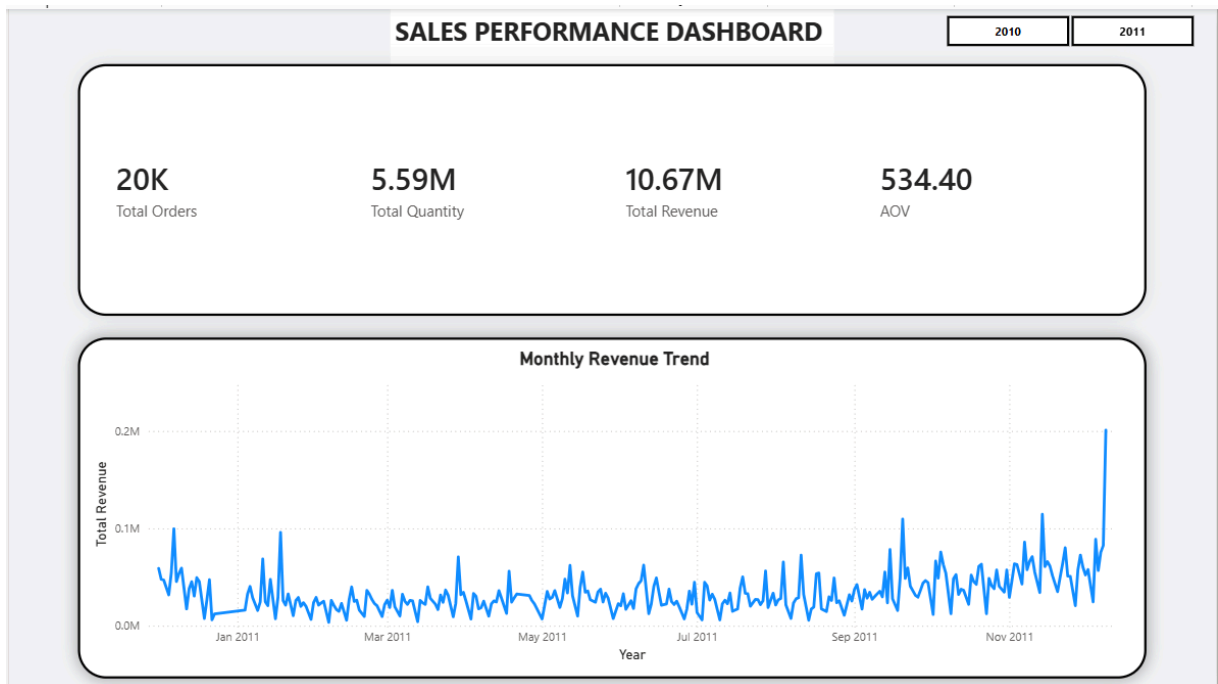
Chapter 5: Dashboard Development Using Power BI

The analytical dashboard was developed in Power BI to provide a comprehensive view of the e-commerce business from three different perspectives: temporal performance, product performance, and regional/customer performance. The dashboard was designed based on the Star Schema data warehouse model, where the central fact table (Fact_Orders) is connected to the dimension tables Dim_Time, Dim_Product, Dim_Customer, and Dim_Region.

The objective of the dashboard is to transform raw transactional data into meaningful business insights, enabling managers to monitor key performance indicators (KPIs), evaluate sales performance, identify top-performing products, and understand customer purchasing behavior across different geographical regions.

I. Revenue Overview Dashboard

The first dashboard page provides a high-level overview of the company's business performance over time.



1.1 Key Performance Indicators (KPIs)

Four KPI cards are displayed at the top of the dashboard to provide an immediate summary of overall business performance:

1.1.1 Total Quantity

- This KPI represents the total number of products sold across all transactions included in the dataset.
- The analysis shows that approximately 6 million units were successfully sold during the observed period. This indicator reflects the operational scale of the business and provides insight into overall sales volume.

1.1.2 Total Orders

- This KPI measures the total number of unique invoices generated by customers.
- The dashboard indicates that the company processed approximately 20,000 orders, demonstrating a significant level of customer activity and transaction frequency.

1.1.3 Total Revenue

- This KPI represents the total sales revenue generated after data cleaning and transformation.
- The total accumulated revenue reached approximately 10.67 million monetary units, making it the most important indicator for evaluating business performance.

1.1.4 Average Order Value (AOV)

Average Order Value measures the average revenue generated per order and is calculated as:

- $AOV = \text{Total Revenue} / \text{Total Orders}$

The calculated AOV is approximately 534.40, indicating the average amount spent by customers in a single transaction.

1.2 Revenue Trend Analysis

A line chart is used to visualize revenue performance across time.

The chart displays revenue fluctuations from late 2010 through 2011 and reveals several important patterns:

- Revenue remained relatively stable during the first half of the year.
- Moderate growth can be observed throughout the middle of the year.
- Significant increases occur during the final quarter.
- The highest revenue peaks appear near the end of the dataset.

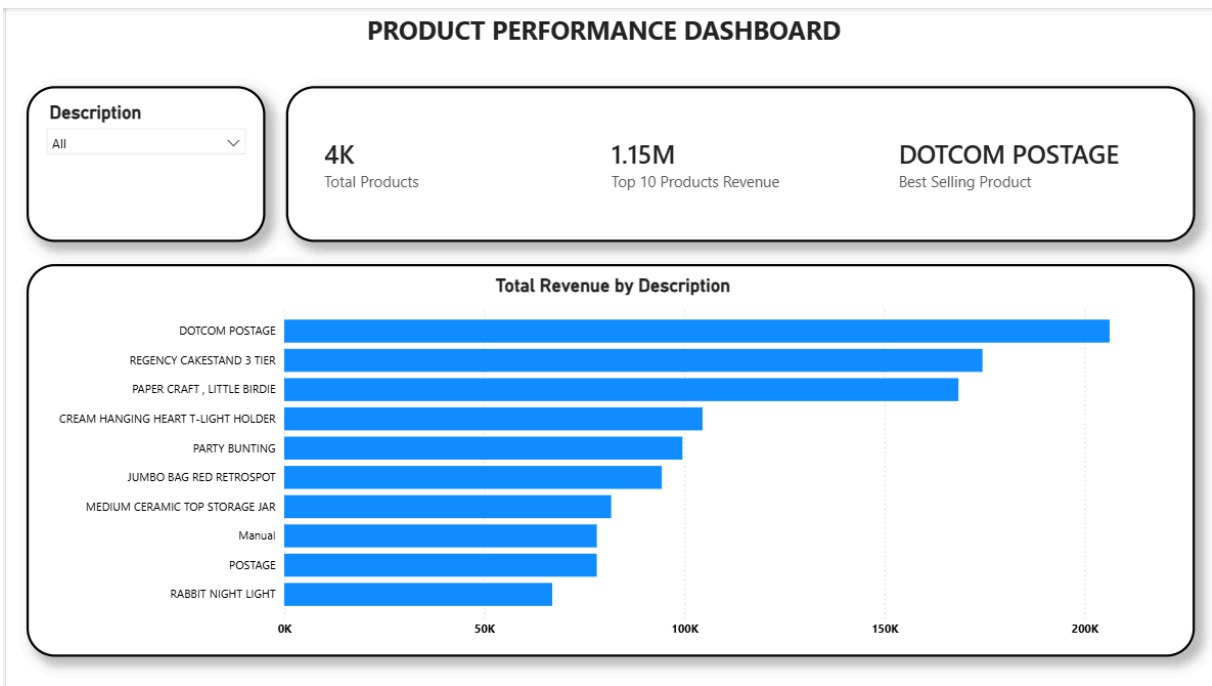
The increasing trend observed in the last months suggests strong seasonal effects, likely associated with holiday shopping periods and increased consumer spending toward year-end.

1.3 Time-Based Filtering

- A year slicer is included to allow users to filter and compare revenue performance between different years.
- This feature enables decision-makers to focus on specific periods and evaluate changes in business performance over time.

II. Product Performance Dashboard

The second dashboard page focuses on product-level analysis and provides insights into which products contribute most significantly to overall revenue. The purpose of this page is to support inventory planning, purchasing decisions, and product portfolio optimization.



2.1 Product Filtering

A product description slicer is provided on the left side of the dashboard. Users can:

- Search for specific products.
- Filter the entire dashboard by product.
- Analyze individual product performance.
- Compare revenue contributions across products.

This interactive feature improves exploration capabilities and allows detailed product-level analysis.

2.2 Product KPI Cards

- Three KPI cards summarize overall product performance.

2.2.1 Total Products

- This KPI displays the total number of unique products stored in the warehouse.
- The dashboard identifies approximately 4,000 distinct products, highlighting the diversity of the company's product catalog.

2.2.2 Revenue of Top 10 Products

- This KPI measures the total revenue generated by the ten highest-performing products in the dataset. Based on the analysis, the top ten products collectively generated approximately 1.15 million in revenue.
- The result demonstrates that a small number of products contribute significantly to overall business performance. Among these products, DOTCOM POSTAGE generated the highest revenue, followed by products such as REGENCY CAKESTAND 3 TIER, PAPER CRAFT, LITTLE BIRDIE, and CREAM HANGING HEART T-LIGHT HOLDER.
- This pattern indicates that revenue is not distributed evenly across all products. Instead, a limited group of high-performing products accounts for a substantial share of total sales. From a business perspective, these products can be considered key revenue drivers and should receive priority in inventory planning, promotional campaigns, and demand forecasting.
- The analysis also supports the Pareto principle, where a relatively small proportion of products contributes a disproportionately large percentage of total revenue. Identifying and monitoring these products enables managers to focus resources on items that have the greatest impact on financial performance.

2.2.3 Best Selling Product

- The dashboard dynamically identifies the product generating the highest revenue.
- Based on the current dataset, the top-performing product is DOTCOM POSTAGE. This product contributes more revenue than any other individual product in the dataset.

2.3 Top Products by Revenue

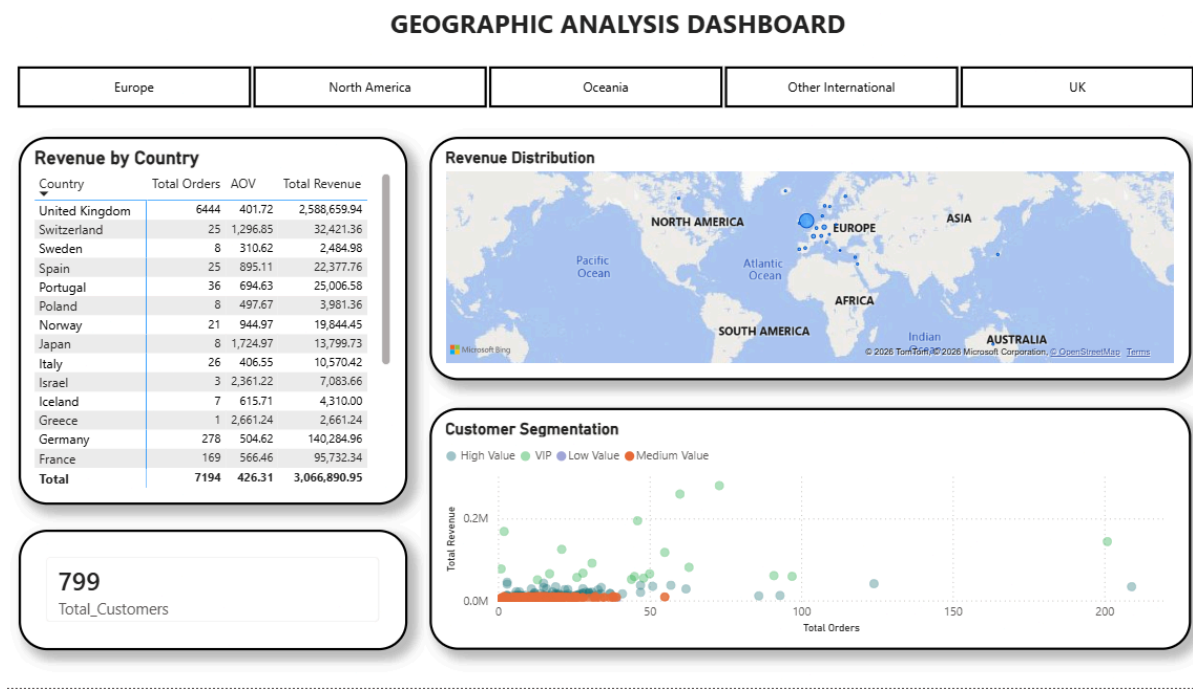
A horizontal bar chart is used to visualize the highest-revenue products. The chart highlights several products with exceptional performance, including:

- DOTCOM POSTAGE
- REGENCY CAKESTAND 3 TIER
- PAPER CRAFT, LITTLE BIRDIE
- CREAM HANGING HEART T-LIGHT HOLDER
- PARTY BUNTING

The visualization enables management to quickly identify products that have the greatest financial impact on the business.

III. Regional and Customer Analysis Dashboard

The third dashboard page focuses on geographical performance and customer behavior. This page supports strategic decision-making regarding market expansion, customer segmentation, and regional performance evaluation.



3.1 Regional Filtering

A Region Group slicer enables users to analyze performance across different geographical areas:

- UK
- Europe
- North America

- Oceania
- Other International

Selecting a region automatically updates all visualizations on the page. This feature allows managers to focus on specific markets and compare regional performance.

3.2 Country Performance Table

A detailed table presents key metrics for each country:

- Total Orders
- Average Order Value
- Total Revenue

The table reveals substantial differences in business performance across countries.

The United Kingdom clearly dominates all metrics, generating significantly higher revenue and order volume compared to other markets. Countries such as the Netherlands, Germany, France, and EIRE also contribute meaningful revenue but at a much smaller scale.

3.3 Geographic Revenue Distribution

A map visualization displays revenue by country using proportional bubbles. Bubble size reflects the magnitude of revenue generated by each market.

The visualization demonstrates that:

- The United Kingdom is the primary revenue source.
- European countries represent the most important international markets.
- Revenue contributions from North America and Oceania are comparatively smaller.

This geographic perspective helps management identify high-value markets and potential expansion opportunities.

3.4 Customer Segmentation Analysis

A scatter plot is used to analyze customer purchasing behavior. The chart plots:

- Total Orders on the X-axis.
- Total Revenue on the Y-axis.

Each point represents an individual customer.

Customers are classified into four segments:

- Low Value
- Medium Value

- High Value
- VIP

Color coding enables quick identification of customer groups. The scatter plot reveals significant variation in customer spending behavior. While many customers generate relatively low revenue, a small number of customers contribute exceptionally high revenue and order volumes.

These customers form the VIP segment and represent the most valuable customer group for the company.

Chapter 6: Business Insights and Findings

I. Strong Seasonal Revenue Patterns

- Analysis of the Revenue Overview Dashboard reveals a clear seasonal pattern in customer purchasing behavior throughout the observed period.
- Although revenue remains relatively stable during most months of the year, a significant upward trend emerges during the final quarter. Revenue growth accelerates noticeably from September onward and reaches its highest level in November and December. The line chart demonstrates increasingly frequent revenue spikes during this period, indicating higher transaction volume and stronger customer demand.
- The dashboard shows that total revenue reached approximately 9.84 million in 2011, generated from more than 18,000 completed orders and over 5.23 million units sold. While Average Order Value (AOV) remained relatively stable at approximately 534.40, both the number of orders and total quantity sold increased substantially.
- This observation suggests that revenue growth was primarily driven by an increase in transaction volume rather than larger individual purchases. Customers were not necessarily spending significantly more per order; instead, more customers were placing more orders and purchasing more products during peak shopping periods.
- The seasonal trend is consistent with typical retail purchasing behavior associated with holiday shopping events, year-end promotions, and Christmas-related demand. Therefore, inventory planning and marketing campaigns should be strengthened several months before the final quarter to maximize sales opportunities.

II. Revenue Concentration Among Top Products

- The Product Performance Dashboard reveals a highly concentrated revenue structure.
- Although the company offers approximately 4,000 unique products, only a small subset contributes a disproportionately large share of total revenue. The KPI "Top 10 Products Revenue" indicates that the ten highest-performing products generated approximately 1.15 million in revenue.

- Among these products, DOTCOM POSTAGE generated the highest revenue and appears as the leading product within the entire product portfolio. Other products such as:
 - + REGENCY CAKESTAND 3 TIER
 - + PAPER CRAFT, LITTLE BIRDIE
 - + CREAM HANGING HEART T-LIGHT HOLDER
 - + PARTY BUNTING
 - + JUMBO BAG RED RETROSPOT
 also contributed significant revenue.
- The bar chart clearly illustrates that revenue is unevenly distributed across products. A relatively small group of products consistently outperforms the rest of the catalog.
- This finding supports the well-known Pareto Principle, which suggests that a relatively small percentage of products often generates a large percentage of total revenue. Consequently, these high-performing products should receive priority in inventory management, procurement planning, promotional campaigns, and demand forecasting activities.

III. Heavy Dependence on the United Kingdom Market

The Geographic Analysis Dashboard demonstrates that the business is heavily dependent on the United Kingdom market.

The country-level revenue table indicates that the United Kingdom generated approximately:

- 18,019 orders
- 9.03 million revenue
- Average Order Value of approximately 500.87

Compared with other countries, the United Kingdom contributes the overwhelming majority of total business revenue.

While countries such as:

- Netherlands
- Germany
- France
- Australia
- Spain

also contribute meaningful revenue, their performance remains significantly lower than that of the UK market.

- The map visualization clearly illustrates this concentration. The largest revenue bubble is located within the United Kingdom region, while other countries are represented by substantially smaller markers.
- From a business perspective, this market dominance provides stability because the company has already established a strong customer base within its domestic market. However, it also creates dependency risk. Economic downturns, regulatory changes, or shifts in consumer behavior within the UK could have a significant impact on overall business performance.
- Therefore, management should consider expanding customer acquisition efforts in international markets to diversify revenue sources and reduce reliance on a single geographic region.

IV. Customer Segmentation and Customer Value Distribution

- Customer segmentation analysis reveals substantial differences in purchasing behavior across customer groups.
- Using the scatter plot, customers were classified into four segments:
 - + Low Value
 - + Medium Value
 - + High Value
 - + VIP
- The majority of customers belong to the Low Value and Medium Value categories. These customers generally place fewer orders and generate relatively limited revenue.
- Conversely, the VIP segment contains a much smaller number of customers but contributes significantly higher revenue and purchase frequency. These customers are represented by points located farther away from the main cluster, particularly in the upper-right region of the scatter plot.
- Several VIP customers generate revenue exceeding **200,000** while simultaneously maintaining a high number of orders. This indicates strong customer loyalty and repeated purchasing behavior.

- The distribution suggests that customer value is highly uneven. Most revenue is generated by a relatively small number of highly engaged customers, while the majority of customers contribute modest revenue amounts.
- This finding highlights the importance of customer retention programs. Maintaining relationships with VIP customers may generate greater long-term value than focusing exclusively on acquiring new customers.

V. Relationship Between Orders and Revenue

- The Customer Segmentation scatter plot also provides insight into the relationship between purchasing frequency and revenue generation.
- A positive correlation can be observed between Total Orders and Total Revenue. In general, customers who place more orders tend to generate higher revenue.
- However, the relationship is not perfectly linear. Several customers achieve relatively high revenue despite having a moderate number of orders. This indicates that certain customers place larger-value orders, resulting in higher Average Order Values.
- At the same time, some customers generate revenue through frequent purchasing activity rather than large individual transactions.
- This distinction suggests the existence of multiple customer purchasing patterns:
 1. High-frequency customers
 2. High-spending customers
 3. VIP customers who combine both characteristics
- Understanding these patterns allows the business to design more targeted marketing strategies and allocate resources more effectively across different customer segments.

Chapter 7: Conclusion and Recommendations

I. Conclusion

This project successfully implemented a complete Business Intelligence solution, including data extraction, transformation, loading, warehouse modeling, and dashboard visualization.

The ETL process cleaned and standardized the raw transactional data before loading it into a PostgreSQL-based data warehouse structured according to Star Schema principles.

Power BI dashboards were then developed to provide analytical views of:

- Revenue performance over time.
- Product performance.
- Regional and customer behavior.

The resulting system enables decision-makers to access accurate, interactive, and data-driven insights.

II. Strategic Recommendations

2.1 Inventory Planning

- The company should increase inventory preparation before the final quarter of the year.
- Historical data indicates a strong increase in demand during year-end shopping periods. Early inventory planning can reduce the risk of stock shortages and lost sales opportunities.

2.2 Product Portfolio Optimization

- Management should prioritize products that consistently generate high revenue. Marketing budgets, promotional campaigns, and inventory investments should focus on the highest-performing products identified in the dashboard.

2.3 Customer Retention Strategy

The customer segmentation analysis revealed that a relatively small group of VIP customers contributes a disproportionately large share of total revenue. Therefore, customer retention should be considered a strategic priority.

Maintaining long-term relationships with high-value customers may generate greater business value than focusing exclusively on acquiring new customers. The company should implement dedicated retention programs targeting VIP customers to strengthen loyalty and encourage repeat purchases.

Recommended initiatives include:

- VIP loyalty programs.
- Personalized promotions and product recommendations.
- Exclusive access to new product launches.
- Premium customer support services.
- Early access to seasonal campaigns and special events.

These initiatives can improve customer satisfaction, increase repeat purchase frequency, enhance Customer Lifetime Value (CLV), and contribute to long-term revenue stability.

2.4 International Market Expansion

Although the United Kingdom remains the primary market, the company should continue expanding its presence in other regions, particularly European markets that already demonstrate strong purchasing activity.

Diversifying revenue sources can reduce dependence on a single market and improve long-term business stability.

2.5 Future BI System Enhancement

Future improvements may include migrating the ETL process to cloud-based platforms such as AWS, Azure, or Google Cloud.

Automated data pipelines and scheduled refresh mechanisms would enable near real-time reporting, allowing management to respond more quickly to market changes and customer behavior trends.